



### Q1. How important is the Python language in today's smart era in the IoT domain?

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**A1.** Python is a general-purpose, high-level programming language, and is currently among the most popular programming languages in the world. Python is used in many applications ranging from developing desktop GUI applications, website applications, and embedded systems to IoT (Internet of Things) applications. It supports both object and structure-oriented language. As compared to other high-level programming languages, Python has simple programming syntax. Hence, it is easy to read and maintain the code.

Python is frequently used by developers as it is more flexible than Java or C programming language. Most developers use Python language in applications using Raspberry

Pi, Intel Edison, NodeMCU, ESP8266, ESP32, Arduino, and other boards, which are very common in IoT applications.

Python has a large number of pre-built libraries in areas such as Web service tools, machine learning, deep learning, etc. Python's clean syntax is suitable for database arrangement and data analysis in IoT systems. It has a large community that can help you with solutions and libraries as and when required.

For further detail on the use of Python in IoT applications, you may refer the following links:

<https://digitalconquerer.com/dev-corner/role-of-python-in-iiot-development/>, <https://techbeacon.com/app-dev-testing/top-6-programming-languages-iiot-projects>, and <https://www.pythontraining.net/usage-of-python-in-iiot>

### Q2. Explain network-on-chip (NoC). What are the advantages of NoC design over system-on-chip (SoC) design?

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**A2.** NoC is a network-based communication subsystem on a semiconductor integrated chip, mostly between SoC modules. SoC is a chip that integrates components of a computer including a central processing unit (CPU), graphics, and memory interfaces. In the SoC designs, pre-designed intellectual property (IP) cores/blocks are becoming an important part. As dimensions of semiconductor chips are shrinking and more IP cores are added to them, many SoC chips are too complex to utilise a traditional data bus or crossbar interconnect approach. The physical resources carrying data on the chips and quality of services begin to crumble.

NoC technology brings notable improvements over conventional data bus and crossbar communication architectures. NoC scheme is to organise communication between operating modules located on the same chip. The scheme improves the scalability and power efficiency of complex SoCs. A typical schematic multiprocessor design with SoC/NoC is shown in Fig. 1.

Some of the advantages of NoC over SoC are given below:

1. NoC provides structured architecture, lowering complexity and cost of chip design
2. Allows reuse of components, architectures, design methods, and tools

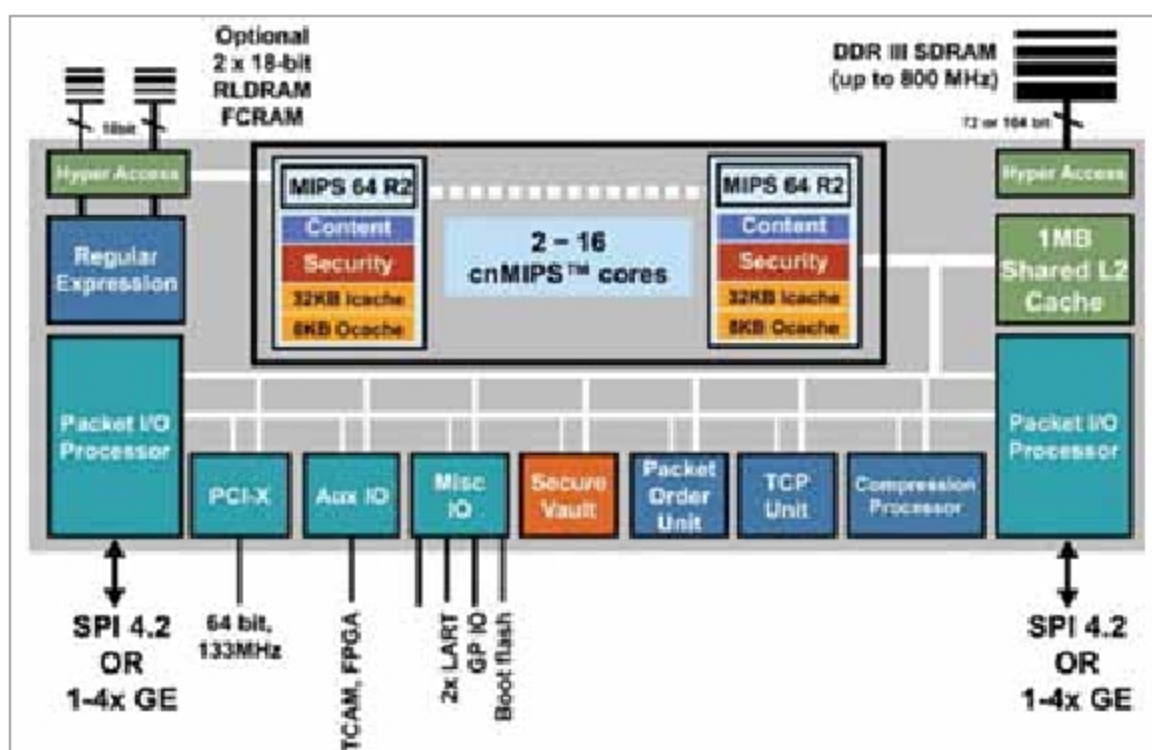


Fig. 1: A typical schematic multiprocessor design with SoC/NoC (Credit: www.sciencedirect.com)

3. Provides efficient and high-performance interconnect
4. Improves scalability of communication architecture
5. Establishes a relationship between the processor blocks by routing the data packets, providing flexible and easy accommodation of changes
6. Reduces wire routing congestion and avoids timing closure issues
7. Simplifies switching and routing functions to achieve higher operating frequencies